1. m/s = 18/5 km/hr
2. km/hr = 5/18 m/s
3. m/min = 60/1000 km/hr
4. Minute hand: 6 degree in 1 min
5. Hour hand = 0.5 degree in 1 min
6. Second Hand = 6 degree in 1 second
7. Hands coincide every 720/11 minutes = 65 5/11 minutes = 65 minutes 27.27 seconds
8. Angle made between H and M hands = |11H-5.5M|
9. Covariance (X, Y) = R (correlation coefficient between X&Y) × S (X) × S (Y)
10. Coefficient of Variation = S/M x 100
11. Men × Days × Work = Men × Days × Work (Person1 x Days1 x Trees1 = Person2 x Days2 x Trees2)
12. Efficiency Ratio: B is 25% more efficient than A => B:125% = 5:4
    Time Ratio => 4:5 (Use 4/5 not 4/9)
13. For two people, time together = ab/(a+b)
14. One-day work = 1/days
15. Work = Rate x Time

    ### Compound Interest (CI) Formula

    If:

    - P = Principal
    - R = Rate of interest per annum (%)
    - T = Time in years

    Then the **Amount** is:

    $$A = P\left(1 + \frac{R}{100}\right)^T$$

    And the **Compound Interest** is:

    $$CI = A - P$$

    or

    $$CI = P\left[\left(1 + \frac{R}{100}\right)^T - 1\right]$$
16.

For 2 years

$$CI - SI = P\left(\frac{R}{100}\right)^2$$

For 3 years

$$CI - SI = P\left(\frac{R}{100}\right)^2\left(3 + \frac{R}{100}\right)$$

If compounded half-yearly

Use:

$$A = P\left(1 + \frac{R/2}{100}\right)^{2T}$$

- Rate becomes $R/2$
- Time becomes $2T$

If compounded quarterly

$$A = P\left(1 + \frac{R/4}{100}\right)^{4T}$$

17.

18.

In traditional logic:
A-type (Universal Affirmative): “All S are P.”
E-type (Universal Negative): “No S are P.”
I-type (Particular Affirmative): “Some S are P.”
O-type (Particular Negative): “Some S are not P.”

Contradictory pairs are:
A ↔ O (All S are P ↔ Some S are not P)
E ↔ I (No S are P ↔ Some S are P)

A. All printers are computer peripherals. → A-type
B. Some computer peripherals are printers. → I-type
C. Some printers are not computer peripherals. → O-type
D. No printers are computer peripherals. → E-type

A ↔ O → (A and C)
They cannot both be true and cannot both be false.

sub-contrary statements are two particular propositions (both can be true but cannot both be false (I-O)
contrary – cannot both be true but can both be false (A-E)

19. Mean = np (no. of trials x probability of success)
20. And Standard Deviation = $\sqrt{n \times p \times (1-p)}$
21. Sum of squares of first n natural numbers = $\frac{n(n+1)(2n+1)}{6}$
22. Sum of first n natural numbers = $\frac{n(n+1)}{2}$
23. Sum of first n odd numbers = n^2
24. Sum of first n even numbers = n(n+1)
25. Sum of cubes of first n natural numbers = $\left(\frac{n(n+1)}{2}\right)^2$

26.

1. $(a + b)^2 = a^2 + 2ab + b^2 = (-a - b)^2$
2. $(a - b)^2 = a^2 - 2ab + b^2$
3. $(a - b)(a + b) = a^2 - b^2$
4. $(a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$
5. $(a + b - c)^2 = a^2 + b^2 + c^2 + 2ab - 2bc - 2ca$
6. $(a - b + c)^2 = a^2 + b^2 + c^2 - 2ab - 2bc + 2ca$
7. $(-a + b + c)^2 = a^2 + b^2 + c^2 - 2ab + 2bc - 2ca$
8. $(a - b - c)^2 = a^2 + b^2 + c^2 - 2ab + 2bc - 2ca$
9. $(a + b)^3 = a^3 + b^3 + 3ab(a + b)$
10. $(a - b)^3 = a^3 - b^3 - 3ab(a - b)$
11. $a^3 + b^3 = (a + b)^3 - 3ab(a + b)$
    $= (a + b)(a^2 - ab + b^2)$
12. $a^3 - b^3 = (a - b)^3 + 3ab(a - b)$
    $= (a - b)(a^2 + ab + b^2)$
13. $a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$

if $a + b + c = 0$ then $a^3 + b^3 + c^3 = 3abc$

Fallacy	Quick Memory Trick
Begging the Question	Conclusion assumed in premise
Ad Hominem	Attack the person
Hasty Generalization	Small sample → big conclusion
Slippery Slope	Small step → disaster
Red Herring	Change the topic
Straw Man	Misrepresent argument
False Cause	Correlation = causation
False Dilemma	Only two choices shown
Appeal to Authority	Expert said it
Bandwagon	Everyone believes it
Appeal to Emotion	Feelings instead of facts
False Analogy	Bad comparison
Loaded Question	Assumes guilt

27.

Ambassador cars have almost disappeared. This is an ambassador car. Therefore, this car has almost disappeared" is Fallacy of Division (B).
The Fallacy of Division occurs when it is assumed that what is true of the whole must be true for the parts.
Hasty generalisation = Fallacy of Composition – what is true for parts is true for whole
Equivocation (A) would involve using a word in different senses within the argument

28. When **equal distances** are covered at two different speeds $x$and y, the average speed is **not** $\frac{x+y}{2}$. The formula is: $\boxed{\text{Average Speed} = \frac{2xy}{x+y}}$

29.

1	Byte	=	8	Bits

1	KB	=	1024	Bytes
1	MB	=	1024	KB
1	GB	=	1024	MB
1	TB	=	1024	GB
1	PB	=	1024	TB
1	MB	=	2^20	Bytes

Formula	Represents
πr^2	Area of a circle
$2\pi r$	Circumference of a circle
$\pi r^2 h$	Volume of a cylinder
$2\pi rh$	Curved surface area of a cylinder
$2\pi r(r+h)$	Total surface area of a cylinder
$\frac{1}{3}\pi r^2 h$	Volume of a cone
$\frac{4}{3}\pi r^3$	Volume of a sphere
$4\pi r^2$	Surface area of a sphere

30.

Area of a rhombus = ½ x D1 x D2

Area of equilateral Triangle = (root3)/4 (side)^2

31.

The Golden Rule

Whenever you have:

```
Lower value
      \
       \
 Desired
       /
      /
Higher value
```

The ratio is always:

$$\boxed{(\text{Higher} - \text{Desired}) : (\text{Desired} - \text{Lower})}$$

Notice the **cross subtraction**.

In alligation:

- If the desired value is **closer to the lower value**, you need **more of the lower quantity**.
- If the desired value is **closer to the higher value**, you need **more of the higher quantity**.

32. Speed downstream: b+s=d/t
33. Speed upstream = b-s = d/t
34. Train length = distance – platform (opposite direction)
35. The decimal number equivalent to binary digit $110 = 1 \times 2^2 + 1 \times 2^1 + 0 \times 2^0$
    $(102)_4 = 1\times4^2+0\times4^1+2\times4^0$
36.

Basic Conditions to test as to when an individual is resident in India:
An individual is said to be a resident in India in any previous year, if he fulfils at least one of the following 2 basic conditions [Section 6(1)]:
Basic condition (a): He is in India in the previous year for a period of 182 days or more.
OR
Basic condition (b): He is in India for a period of 60 days or more during the previous year AND 365 days or more during the 4 years immediately preceding the previous year.

Bret Lee satisfies the second basic condition because he is in India for more than 60 days during the relevant previous year and for 400 days during four years preceding the relevant previous year. Therefore, **he is a resident.**

Additional Conditions to test as to when a resident individual is Ordinarily Resident in India:
A resident individual is treated as resident and ordinarily resident in India if he satisfies both of the 2 additional conditions:
Additional Condition 1: He has been resident in India in at least 2 out of 10 previous years immediately preceding the relevant previous year;
AND
Additional Condition 2: He has been in India for a period of 730 days or more during the 7 years immediately preceding the relevant previous year.

Therefore, it can be said that an individual becomes resident and ordinarily resident in India, if he satisfies at least one of the basic conditions (i.e. (a) or (b)) and the two additional conditions (i.e., 1 and 2). In case resident satisfies any one of the above conditions, he is said to be a resident but not ordinarily resident in India (RNOR).

In the given case of Bret Lee, although he satisfies the first additional condition of being resident for at least 2 out of 10 preceding previous years but he does not satisfy the second additional condition as during 7 years preceding the previous year, he is in India for only 700 days. He shall, therefore, be a **resident but not ordinarily resident in India (RNOR).**

Therefore, the correct answer is option (C).

33. T-test: parametric – means of 2 groups – normal distribution

F-test: parametric – variance of 2 groups – normal distribution

U-test: non parametric – Mann-Whitney U test – when t-test fails

H-test – Non parametric – Kruskal Wallis test – when assumptions of ANNOVA are not met

Time and Work

1. LCM
2. MDH/W = MDH/W

Boats and Streams

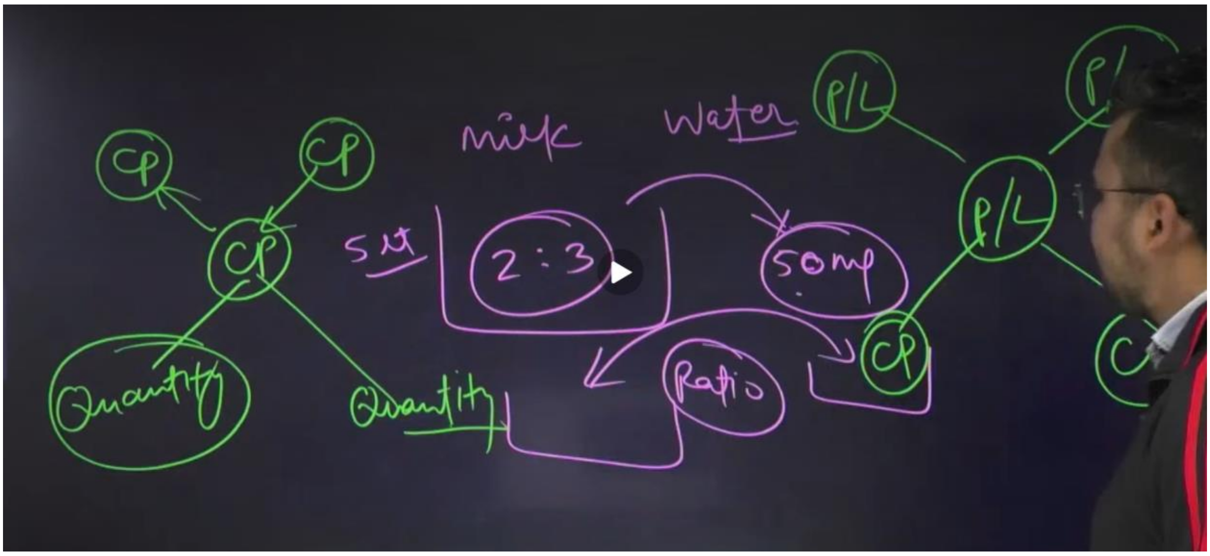
Speed of the boat = x

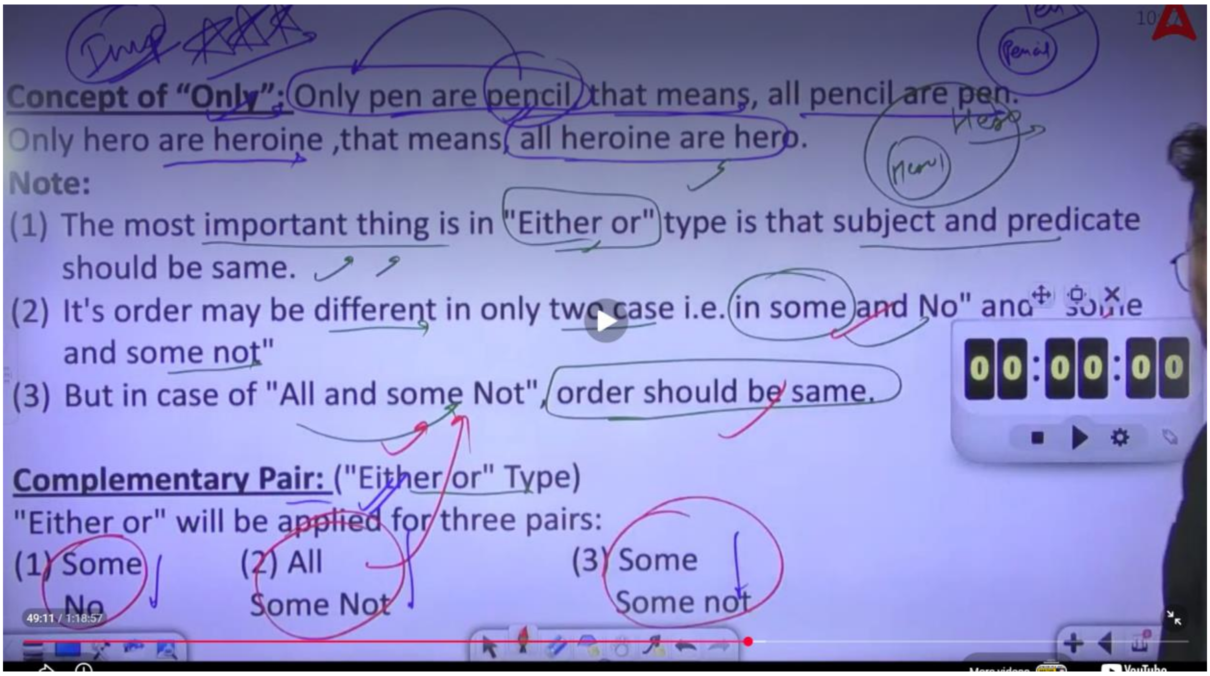
Speed of the stream = y

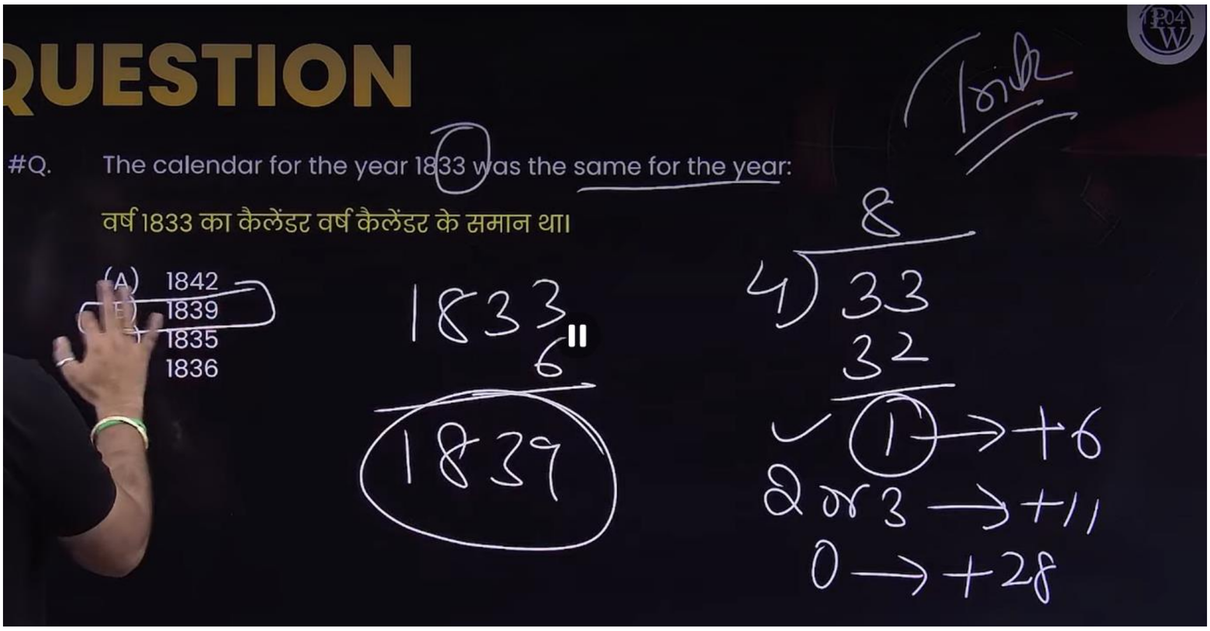
Downstream = x+y

Upstream = x-y

Mixtures and Allegations

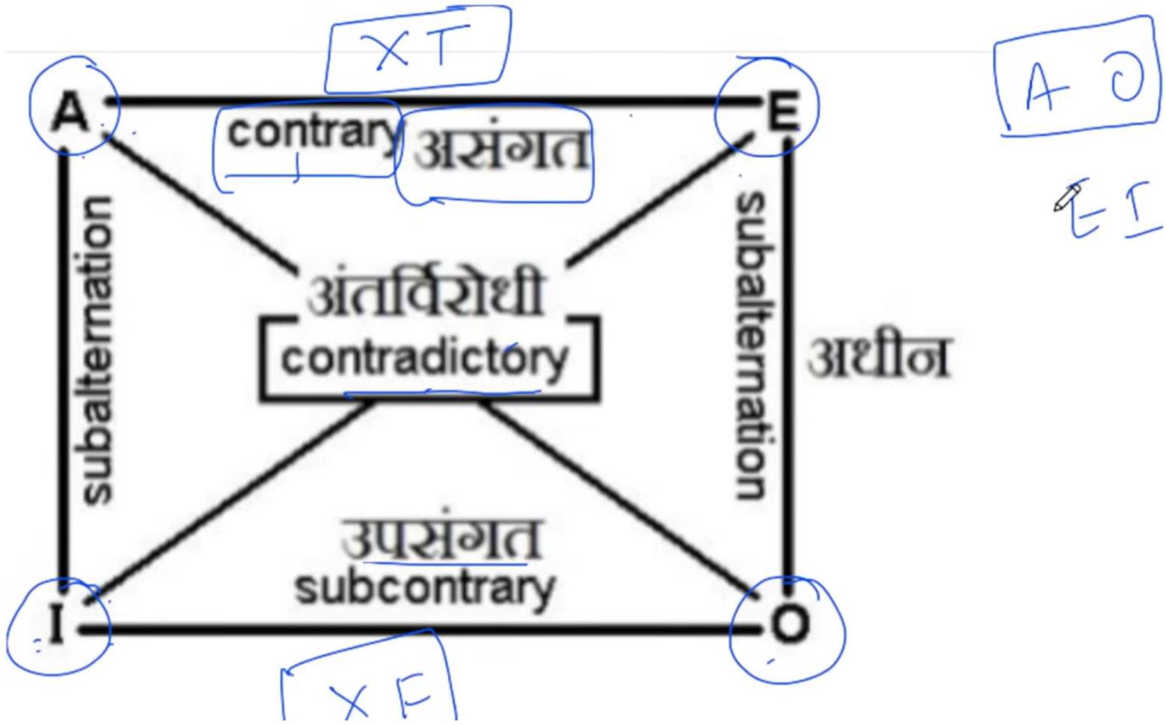

milk
water
P/L
CP
2 : 3
Ratio
Quantity

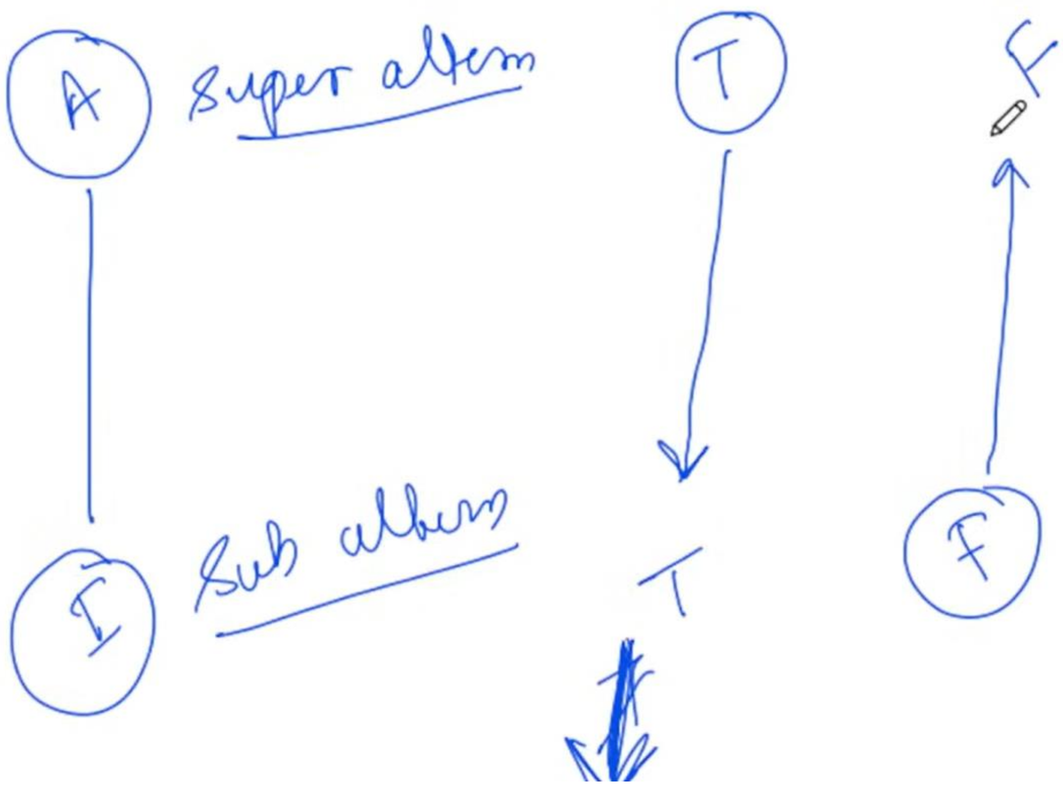

Concept of "Only": Only pen are pencil that means, all pencil are pen.
Only hero are heroine ,that means, all heroine are hero.
Note:
(1) The most important thing is in "Either or" type is that subject and predicate should be same.
(2) It's order may be different in only two case i.e. in some and No" and "some and some not"
(3) But in case of "All and some Not", order should be same.
Complementary Pair: ("Either or" Type)
"Either or" will be applied for three pairs:
(1) Some
No
(2) All
Some Not
(3) Some
Some not
Pencil
Hero
Heroine
00:00:00
49:11 / 1:18:57



Normal Year = 365 Days, Odd Days = 1, 1st day of the year = last day of the year

Leap year = 366 Days, Odd Days = 2, 1st day of the year = 2nd last day of the year

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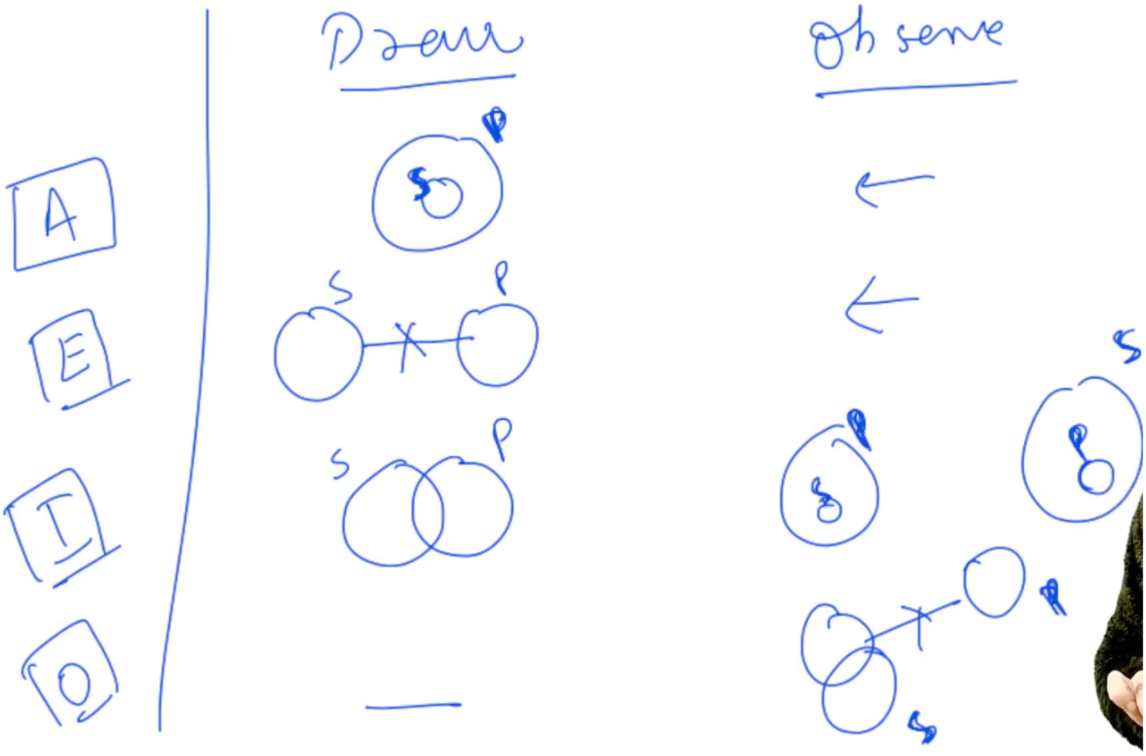
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Pramanas in Different School of Thought

Astik

Vedanta - Pratyakṣa, Anumāna, Upamāna, Śabda, Arthpatti, Anupalabdhi
Nyaya - Pratyakṣa, Anumāna, Upamāna, Śabda
Vaisheshika - Pratyaksa and Anumana
Minimasa - Pratyaksa, Anumana and Shabda
Sankhya - Pratyaksa, Anumana and Shabda

Nastik

Buddhist – Pratyaksh, Anumana
Jainism - Pratyaksh, Anumana, Shabd
Charwak - Pratyaksh

Quick Recap

Numerical Value	*Terminology*	*Description*	*Shape of the Demand curve*
$e_p = \infty$	Perfectly elastic	Change in demand is infinite at a given price	Horizontal
$e_p = 0$	Perfectly inelastic	Demand remains unchanged whatever be the change in price	Vertical
$e_p = 1$	Unitary elastic	$\%\Delta Q = \%\Delta P$	Rectangular Hyperbola
$0 < e_p < 1$	Inelastic	$\%\Delta Q < \%\Delta P$	Steeper
$\infty > e_p > 1$	Elastic	$\%\Delta Q > \%\Delta P$	Flatter

Price Elasticity of Demand

$$E_d = \frac{\%\Delta Q}{\%\Delta P}$$

Symbol	Meaning
E_d	Price Elasticity of Demand
ΔQ	Change in Quantity Demanded
ΔP	Change in Price
%	Percentage Change

Income Elasticity

$$E_y = \frac{\%\Delta Q}{\%\Delta Y}$$

Symbol	Meaning
E_y	Income Elasticity
Y	Income
ΔY	Change in Income
Q	Quantity Demanded

Cross Elasticity

$$E_{xy} = \frac{\%\Delta Q_x}{\%\Delta P_y}$$

Symbol	Meaning
Q_x	Quantity of Product X
P_y	Price of Product Y

Revenue

Total Revenue

$$TR = P \times Q$$

Symbol	Meaning
TR	Total Revenue
P	Price per Unit
Q	Quantity Sold

Average Revenue

$$AR = \frac{TR}{Q}$$

Symbol	Meaning
AR	Average Revenue

Marginal Revenue

$$MR = \frac{\Delta TR}{\Delta Q}$$

Symbol	Meaning
MR	Marginal Revenue
ΔTR	Change in Total Revenue
ΔQ	Change in Quantity Sold

Cost

Total Cost

$$TC = TFC + TVC$$

Symbol	Meaning
TC	Total Cost
TFC	Total Fixed Cost
TVC	Total Variable Cost

Average Cost

$$AC = \frac{TC}{Q}$$

Symbol	Meaning
AC	Average Cost
Q	Quantity Produced

Marginal Cost

$$MC = \frac{\Delta TC}{\Delta Q}$$

Symbol	Meaning
MC	Marginal Cost
ΔTC	Change in Total Cost

Profit

$$\pi = TR - TC$$

Symbol	Meaning
π	Profit
TR	Total Revenue
TC	Total Cost

National Income

GDP

$$GDP = C + I + G + (X - M)$$

Symbol	Meaning
C	Consumption Expenditure
I	Investment Expenditure
G	Government Expenditure
X	Exports
M	Imports

GNP

$$GNP = GDP + NFIA$$

Symbol	Meaning
NFIA	Net Factor Income from Abroad

NDP

$$NDP = GDP - Depreciation$$

Symbol	Meaning
Depreciation	Consumption of Fixed Capital

Consumption & Saving

Consumption Function

$$C = a + bY$$

Symbol	Meaning
C	Consumption
a	Autonomous Consumption
b	MPC
Y	Income

Saving Function

$$S = -a + (1 - b)Y$$

Symbol	Meaning
S	Saving
a	Autonomous Consumption
b	MPC

MPC & MPS

MPC

$$MPC = \frac{\Delta C}{\Delta Y}$$

Symbol	Meaning
MPC	Marginal Propensity to Consume
ΔC	Change in Consumption
ΔY	Change in Income

MPS

$$MPS = \frac{\Delta S}{\Delta Y}$$

Symbol	Meaning
MPS	Marginal Propensity to Save
ΔS	Change in Saving

Important Relation

$$MPC + MPS = 1$$

Multiplier

$$K = \frac{1}{1 - MPC}$$

or

$$K = \frac{1}{MPS}$$

Symbol	Meaning
K	Multiplier
MPC	Marginal Propensity to Consume
MPS	Marginal Propensity to Save

Income Change

$$\Delta Y = K \times \Delta I$$

Symbol	Meaning
ΔY	Change in National Income
K	Multiplier
ΔI	Change in Investment

Money & Banking

Money Multiplier

$$m = \frac{1}{RR}$$

Symbol	Meaning
m	Money Multiplier
RR	Reserve Ratio

1. Total Revenue (TR)

$$TR = P \times Q$$

or

$$TR = AR \times Q$$

Symbols

- TR = Total Revenue
- P = Price per unit
- Q = Quantity sold
- AR = Average Revenue

---

2. Average Revenue (AR)

$$AR = \frac{TR}{Q}$$

Since:

$$TR = P \times Q$$

$$AR = \frac{P \times Q}{Q}$$

$$AR = P$$

Important NET Fact

$$\boxed{AR = Price}$$

This is asked directly in MCQs.

3. Marginal Revenue (MR)

$$MR = \frac{\Delta TR}{\Delta Q}$$

Symbols

- MR = Marginal Revenue
- ΔTR = Change in Total Revenue
- ΔQ = Change in Quantity Sold

4. Relationship Between AR and MR

Perfect Competition

$$AR = MR = P$$

This is a favorite UGC NET conceptual question.

Monopoly / Imperfect Competition

$$MR < AR$$

Always.

5. TR-MR Relationship

When:

MR > 0

TR increases.

MR = 0

TR is maximum.

MR < 0

TR decreases.

This is asked repeatedly in UGC NET.

6. Linear Demand Relationship

If:

$$AR = a - bQ$$

then

$$MR = a - 2bQ$$

Shortcut

MR has the same intercept but double the slope of AR.

Example:

$$AR = 100 - 2Q$$

Then:

$$MR = 100 - 4Q$$

Very common numerical.

7. Profit Maximization

$$MR = MC$$

where:

- MR = Marginal Revenue
- MC = Marginal Cost

This is probably the single most important microeconomics formula for NET.

Must-Memorize Revenue Formulas

$$TR = P \times Q$$

$$AR = \frac{TR}{Q}$$

$$AR = P$$

$$MR = \frac{\Delta TR}{\Delta Q}$$

$$AR = MR = P \quad \text{(Perfect Competition)}$$

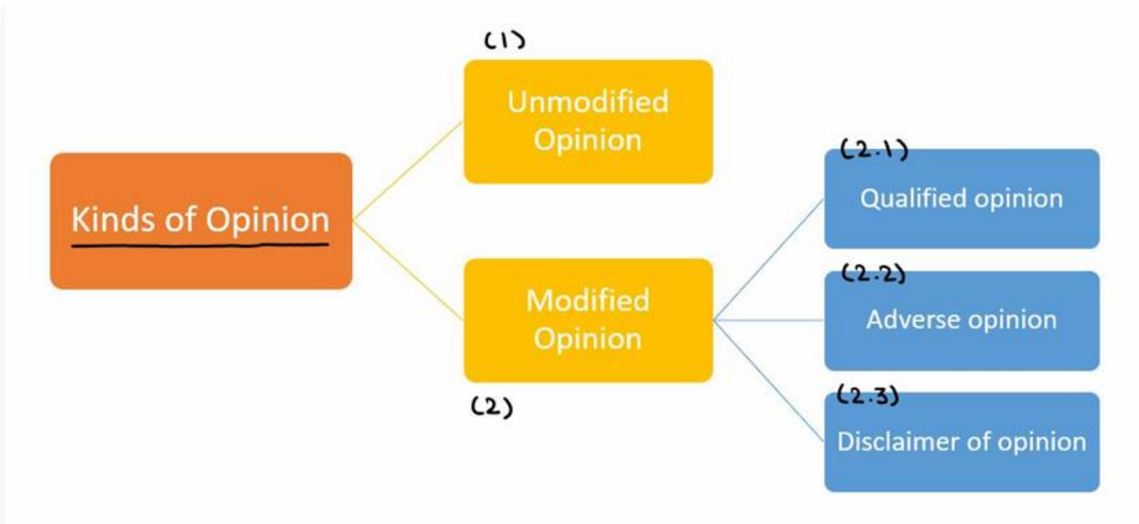
$$MR < AR \quad \text{(Monopoly)}$$

$$MR = MC \quad \text{(Profit Maximization)}$$

$$AR = a - bQ$$

$$MR = a - 2bQ$$

2. Audit Report



1) Unmodified Opinion

It is issued for any audit where the auditor is satisfied that the financial statements present a true and fair view of the operations and transactions in an enterprise during the period. The auditor shall express an unmodified opinion when the auditor concludes that the financial statements are prepared, in all material respects, in accordance with the applicable financial reporting framework.

An audit report with an Unmodified Opinion is also known as a 'Clean Report'. An Unmodified report develops confidence among users of Financial statements and annual reports of an enterprise. It provides an impression that the financial statements are reasonably free from any misstatements and results as appearing there are true and fair.

2) Modified Opinion

A Modified Opinion is issued in an audit report, if the auditor concludes that, based on the audit evidence, the financial statements as a whole are not free from material misstatements; or is unable to obtain sufficient appropriate audit evidence to conclude that the financial statements as a whole are free from material misstatements.

There are 3 kinds of modified opinions which are issued according to the findings and circumstances:
a) **A qualified opinion** — the auditor concludes that, except for specific matters explained in the audit report, the financial statements give a true and fair view;
b) **An adverse opinion** — the auditor concludes that the financial statements do not give a true and fair view;
c) **A disclaimer of opinion** — the auditor concludes that the extent of their inability to obtain sufficient appropriate audit evidence is such that it is not possible to form an opinion on the financial statements.